

CUBAN NATIONAL CHEMISTRY CONTEST

“To work in the sterile when one can do the useful; to deal with the easy when one has spirit to try the difficult, is to strip the dignity from the talent.”

José Martí

Name(s): _____ Last name(s): _____
 School: _____
 Municipality: _____ Province: _____
 CODE: _____ Test: **DAY 2. ELEVENTH GRADE EXAM**

Instructions

- Write clearly all your data in the appropriate place (names, last names, school, municipality, province). Do not write anything in the field CODE.
- Verify that your questions booklet is complete. This is the Day 2 problem set, with an individual exam for 11th grade. The exam has 3 problems (problem 8, 4 items; problem 9, 6 items; problem 10, 10 items) and a total of 5 pages.
- All data, constants, conversions and basic equations needed are included in the exam. You are allowed to use a scientific calculator.
- At the beginning of each problem its scoring (from a total of 100 points), each item scoring (in marks) is indicated.

Success!



Ministerio de Educación
República de Cuba



CONSTANTS, CONVERSIONS AND EQUATIONS

Avogadro's Constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Ideal gas equation	$p \cdot V = n \cdot R \cdot T$
Gasses' constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	Thermodynamics First Principle	$Q = \Delta E + W$
Cero in the Celsius scale	273,15 K	Gibbs' energy	$\Delta G = \Delta H - T \cdot \Delta S$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{C_{\text{red}}}{C_{\text{ox}}}$
Water ionic product at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Arrhenius' equation	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$
Standard conditions	$p^\circ = 100 \text{ kPa}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$	$\Delta G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$	
$1 \text{ J} = 1 \text{ C} \cdot 1 \text{ V} = 1 \text{ kPa} \cdot 1 \text{ L} = 1 \text{ W} \cdot 1 \text{ s}$		$1 \text{ C} = 1 \text{ A} \cdot 1 \text{ s}$	

PERIODIC TABLE OF THE ELEMENTS

																	13 3A		14 4A	15 5A	16 6A	17 7A	18 8A	
1 H 1.008	2 2A																5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18		
3 Li 6.941	4 Be 9.012											13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95							
11 Na 22.99	12 Mg 24.31	3 3B	4 4B	5 5B	6 6B	7 7B	8 8B	9 8B	10 8B	11 1B	12 2B	31 Ga 69.72	32 Ge 72.61	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80							
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.88	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.39	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3							
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po (209)	85 At (210)	86 Rn (222)							
55 Cs 132.9	56 Ba 137.3	57 La 138.9	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	114 Uuq (277)		116 Uuh (277)		118 Uuo (277)								
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (263)	107 Bh (262)	108 Hs (265)	109 Mt (266)	110 Ds (269)	111 Rg (272)	112 Uub (277)													

Problem 8. Molecular equilibrium (15 points)

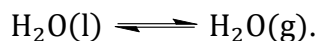
8.1	8.2	8.3	8.4	Total	Total
8	12	3	7	30	15

If you prepare blue crystals of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ to form an aqueous solution of copper(II) sulfate and store them in an open container inside a hot and dry room, the edges will become light blue or white colored, they will “weather” slowly. If you want to keep the blue color of the crystals, you must cover them with a transparent lacquer. There are several copper(II) sulfate hydrates $\text{CuSO}_4 \cdot n\text{H}_2\text{O}$ ($n = 1, 3, 5$), which interconvert depending on the content of water vapor in the air.

The amount of water in the air is usually reported as relative humidity H_r :

$$H_r(\text{in } \%) = \frac{p(\text{H}_2\text{O})}{p^\circ(\text{H}_2\text{O})} \cdot 100$$

Where $p(\text{H}_2\text{O})$ is the vapor pressure of water and $p^\circ(\text{H}_2\text{O})$ is the pressure when the air is saturated with water vapor (when the relative humidity of the air is 100%) at the same temperature. $p^\circ(\text{H}_2\text{O})$ is, by definition, the pressure of water vapor when the following equilibrium is established



8.1. Calculate the water pressure $p^\circ(\text{H}_2\text{O})$ at 25 °C.

8.2. Calculate the value of the relative humidity of the air (in %) at which copper(II) sulfate hydrates with $n = 3$ and $n = 5$ interconvert at 25 °C.

8.3. In which form will you find a crystal of copper(II) sulfate in Cuba in winter? Typical weather conditions in this season after the passage of a cold front are 25 °C temperature and 50% relative humidity.

8.4. Calculate the boiling point of water at the atmospheric pressure. Consider that the atmospheric pressure is equal to the standard pressure ($p = p^\circ = 100 \text{ kPa}$).

Compound	$\Delta H_f^\circ(298 \text{ K})/\text{kJ} \cdot \text{mol}^{-1}$	$S^\circ(298 \text{ K})/\text{J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$
$\text{CuSO}_4 \cdot 5\text{H}_2\text{O} (\text{s})$	-2278,0	305,4
$\text{CuSO}_4 \cdot 3\text{H}_2\text{O} (\text{s})$	-1683,1	225,1
$\text{CuSO}_4 (\text{s})$	-771,4	109,2
$\text{H}_2\text{O} (\text{l})$	-285,8	70,1
$\text{H}_2\text{O} (\text{g})$	-241,8	188,7

Problem 9. Acid-base equilibrium (15 points)

9.1	9.2	9.3	9.4	9.5	9.6	Total	Total
8	3	3	5	3	8	30	15

In contrast with pure water, which is neutral, rainwater is slightly acidic. Some causes for this phenomenon are natural and others are triggered by human action. Natural rainwater contains dissolved carbon dioxide in the air.

9.1. Calculate the pH of natural rainwater. The air contains about 0,035% (v/v) of carbon dioxide and the average atmospheric pressure is 101,3 kPa. $K_{a1}(\text{CO}_2) = 4,47 \cdot 10^{-7}$; $K_{a2}(\text{CO}_2) = 4,68 \cdot 10^{-11}$
 $\text{CO}_2(\text{g}) \rightleftharpoons \text{CO}_2(\text{ac}) \quad K_H = 3,36 \cdot 10^{-7} \text{ mol} \cdot \text{L}^{-1} \cdot \text{Pa}^{-1}$

In the air, sulfur dioxide and nitrogen monoxide waste produced by industrial activity are oxidized to sulfur trioxide and nitrogen dioxide, respectively. These gasses react with water to form sulfuric and nitric acids. These reactions provoke that the so-called acid rain has an average pH value of about 4,5.

9.2. Write the chemical equation for the reaction between nitrogen dioxide and water.

9.3. Calculate which concentration of nitric acid causes a pH of 4,5.

9.4. Calculate which concentration of sulfuric acid ($K_{a2} = 1,02 \cdot 10^{-2}$) causes a pH of 4,5.

Some caves are formed when the water filtered through the fractures (which is enriched in CO_2 content) slowly dissolves the rock in a process that can last for millions of years. This process also creates rock formations such as stalactites and stalagmites. These caves are formed in calcareous massifs of limestone and dolomites. The main components of the rocks are calcite (CaCO_3) and dolomite ($\text{CaMg}(\text{CO}_3)_2$), respectively. The most extensive cave system of this kind is Mammoth Cave, in the United States, with more than 590 km of interconnected galleries. Santo Tomás cave system, in Sierra de los Órganos, in Pinar del Río province, is the largest from Cuba, with an extension of 46,2 km and is also formed by such caves. Its exploration started in 1954 by the Cuban Speleological Society under the direction of the eminent Cuban scientist, patriot and politician Antonio Núñez Jiménez (1923-1998).

9.5. Write the chemical equation for the reaction between carbon dioxide dissolved in natural rainwater and dolomite.

9.6. Calculate the solubility (in $\text{mol} \cdot \text{L}^{-1}$) of calcite in natural rainwater. Assume that the natural rainwater has a pH equal to 5,65. $K_{ps}(\text{CaCO}_3) = 3,36 \cdot 10^{-9}$.

Problem 10. Electrolysis (20 points)

10.1	10.2	10.3	10.4	10.5	10.6	10.7	10.8	10.9	10.10	Total	Total
6	4	2	2	3	8	4	5	4	2	40	20

Chlorine can be prepared in the chemistry laboratory by reacting manganese(IV) oxide with hydrochloric acid.

10.1. Write the chemical equation corresponding to the previous reaction and calculate its equilibrium constant.

Chlorine is obtained at an industrial scale by the electrolysis of sodium chloride in an aqueous solution in a so-called chlor-alkali process. Three methods are employed: electrolysis with a cell with mercury amalgam, electrolysis with a diaphragm cell and electrolysis with a membrane cell.

In Cuba, chlorine is produced in the factory “Elpidio Sosa” from the Electrochemical Enterprise of Sagua la Grande, a city located in the north of Villa Clara province, using the methods of electrolysis with a mercury cathode cell. First, electrolysis of sodium chloride originates in the cathode sodium amalgam Na(Hg) that is later decomposed by reacting with water.

10.2. Write the chemical equations corresponding to the reactions described.

10.3. In your opinion, what is the main drawback of this method?

Electrolysis with a diaphragm cell is mainly employed in North America. In this method, the anode and the cathode are separated to avoid the combination of the gases generated in both compartments. The total reaction that happens is



10.4. Write the cathode and anode half-reactions.

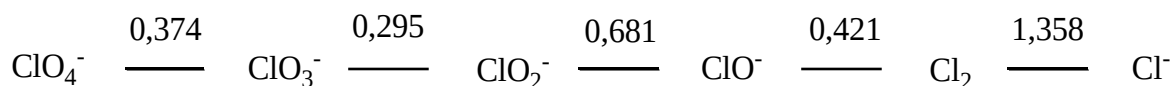
10.5. Calculate the concentration of dissolved sodium chloride. Suppose that it is employed a saturated solution of this salt in water (26,0% in mass, pH = 7, T = 25 °C, density 1,20 g·mL⁻¹).

10.6. Calculate the minimum necessary voltage to carry out the process.

10.7. Write the equations corresponding to other half-reactions that might happen in the cathode and the anode. Comment on why they don't happen.

10.8. Which is the minimum amount of energy (in kW·h) which is needed to obtain 1,42 kg of chlorine (M = 70,9 g·mol⁻¹)? Suppose an efficiency of 100%.

Given the potentials diagram of chlorine in basic media



10.9. Which chlorine species disproportionate in basic media?

10.10. Sodium hypochlorite is employed for the potabilization of water. Write one chemical equation for the obtention of this substance.

Data: E°(acid media)/V: MnO₂/Mn²⁺ = 1,23; O₂/H₂O = 1,23; H⁺/H₂ = 0